Now shrink everything: let B, D, G approach A. The intersection J of BG and AG races toward A so that GJ becomes less than any assignable length, whence $AG : Ag$ differs from equality by nothing in the limit. With $AB^2 = AG \times BD$ and $Ab^2 = Ag \times bd$, the ratio $AB^2 : Ab^2$ is compounded of $AG : Ag$ and $BD : bd$; since the first factor becomes unity, by Lemma I the ultimate ratio $AB^2 : Ab^2$ equals $BD : bd$. Thus the subtense goes as the *square* of the arc — every smooth bend, circle or parabola alike, peels from its tangent quadratically, the universal signature of finite curvature. *Q.E.D.*